

Day 26 — TNPSC Group 2A Study Material

Part A – General Studies: Constitution of India & the Preamble

READING MATERIAL

First of ~12 Unit IV topics. Real exam evidence shows TNPSC favors the Preamble's AMENDMENT history — which words were added when, and which Supreme Court cases established key doctrines — over simply reciting the Preamble's text.

Q: When was the Constitution of India adopted, and when did it come into force?

A: Adopted 26 November 1949 (celebrated as Constitution Day / Samvidhan Divas); came into force 26 January 1950 (celebrated as Republic Day).

Q: What are the exact descriptive words in the Preamble describing India, and what is the source of the ideals of Liberty, Equality, and Fraternity?

A: India is described as a SOVEREIGN, SOCIALIST, SECULAR, DEMOCRATIC REPUBLIC. The ideals of Liberty, Equality, and Fraternity embodied in the Preamble were adopted under the inspiration of the French Revolution (confirmed via a real TNPSC question).

Q: Real exam: When did the Supreme Court first hold 'Secularism' to be a basic feature of the Constitution, and when was the word actually inserted into the Preamble's text?

A: The Supreme Court held secularism to be a basic feature in 1973 (via the Kesavananda Bharati case, which established the wider 'Basic Structure' doctrine). The word 'SECULAR' was then formally inserted into the Preamble's text later, by the 42nd Constitutional Amendment Act, 1976.

Q: What other words did the 42nd Amendment (1976) add to the Preamble, besides 'Secular'?

A: 'SOCIALIST' and 'INTEGRITY' (making 'Unity and Integrity of the Nation') were also added by the same 42nd Amendment — none of these three words were in the ORIGINAL 1949 Preamble.

Q: Real exam: In which case did the Supreme Court hold that 'the Preamble is an integral part of the Indian Constitution'?

A: The LIC of India case (1995) — per a real TNPSC exam question (note: the Kesavananda Bharati case, 1973, is the more famous/foundational ruling establishing that the Preamble is part of the Constitution and can itself be amended, overruling the earlier Berubari Union case of 1960 which had said otherwise; the LIC case appears to reaffirm this in a real exam's specific framing).

Q: What are the THREE dimensions of 'Justice' named in the Preamble?

A: Social, Economic, and Political justice.

Q: What does "WE, THE PEOPLE OF INDIA" at the start of the Preamble signify?

A: That the Constitution derives its authority directly from the people of India, not from any external authority (like the British Crown) or from the states — it establishes the people as the ultimate source of sovereignty.

MOCK TEST (WITH ANSWERS)

1. The Constitution of India was adopted on 26 November 1949 and came into force on:

- A. 15 August 1947
- B. 26 January 1950 ✓**
- C. 2 October 1950
- D. 1 January 1950

Explanation: 26 January 1950 (Republic Day). [easy, authored]

2. The ideals of Liberty, Equality, and Fraternity in the Preamble were adopted under the inspiration of:

- A. The American Declaration of Independence
- B. The French Revolution ✓**
- C. The Russian Revolution
- D. The UN Charter

Explanation: The French Revolution. [medium, question-bank-2017]

3. The Supreme Court first held 'Secularism' to be a basic feature of the Constitution in which year?

- A. 1950

B. 1973 ✓

C. 1976

D. 1995

Explanation: 1973 (Kesavananda Bharati case). [hard, question-bank-2017]

4. The words 'Socialist', 'Secular', and 'Integrity' were added to the Preamble by which amendment?

A. 1st Amendment

B. 26th Amendment

C. 42nd Amendment ✓

D. 44th Amendment

Explanation: 42nd Constitutional Amendment Act, 1976. [medium, authored]

5. In which case did the Supreme Court hold that 'the Preamble is an integral part of the Indian Constitution'?

A. Shankari Prasad case (1951)

B. LIC of India case (1995) ✓

C. Indira Nehru Gandhi case (1975)

D. Maneka Gandhi case (1978)

Explanation: LIC of India case (1995). [hard, question-bank-2025]

6. The Preamble names three dimensions of 'Justice'. Which of the following is NOT one of them?

A. Social

B. Economic

C. Political

D. Religious ✓

Explanation: The three are Social, Economic, and Political — not Religious. [easy, authored]

7. Was the word 'Socialist' present in the Preamble's ORIGINAL 1949 text?

A. Yes, from the start

B. No, added only in 1976 ✓

C. No, added only in 1950

D. Yes, but removed in 1976

Explanation: No — added by the 42nd Amendment in 1976. [medium, authored]

Part B – Aptitude & Mental Ability: Time and Work

READING MATERIAL

Time and Work is the last of the 10 named Aptitude skills — after this, Part B moves to its Reasoning unit. Real exam evidence shows heavy use of 'trap' questions designed to look like they need complex math but actually resolve via a simple ratio.

Q: Method — inverse proportion (more workers, less time): 12 men complete a work in 36 days. In how many days will 18 men finish the same work?

A: Men $\times$ Days = constant (total work). $12 \times 36 = 18 \times x \rightarrow x = 432/18 = 24$ days. More workers means proportionally fewer days.

Q: Method — sequential combined work: A, B, C can complete a job alone in 8, 12, 16 days respectively. A and C work together for 2 days, then C leaves and B joins A. In how many more days do A and B finish?

A: Work rates: A=1/8, B=1/12, C=1/16 per day. In 2 days, A+C complete $2 \times (1/8 + 1/16) = 2 \times (3/16) = 3/8$ of the work. Remaining = 5/8. A+B's combined rate = $1/8 + 1/12 = 5/24$ per day. Days needed = $(5/8) \div (5/24) = (5/8) \times (24/5) = 3$ days.

Q: Method — wages split by share of work done: A can do a job in 10 days, B in 15 days. They work together for 5 days, then C finishes the remaining work in 2 days. Total payment for the whole job is Rs.5,000 — how should it be split?

A: Pay is split in proportion to how much of the TOTAL work each person actually did, not how many days they worked. In 5 days, A completes $5/10 = 1/2$ of the job, B completes $5/15 = 1/3$; together that's $5/6$, leaving $1/6$ for C to finish. Split of Rs.5,000 by work share: A gets $1/2 \times 5000 = \text{Rs.}2,500$; B gets $1/3 \times 5000 \approx \text{Rs.}1,667$; C gets $1/6 \times 5000 \approx \text{Rs.}833$. The key insight is the split follows WORK DONE, not days worked — C worked fewer days than B but did a comparable or different share depending on the rates, and pay tracks output, not time.

Q: Method — pairwise combined rates $\rightarrow$ solve for all three together: A+B finish a job in 12 days, B+C in 15 days, C+A in 20 days. In how many days can A, B, C finish it together?

A: Add all three pairwise rates: $(A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 2(A+B+C)$'s rate. LCM=60: $5/60 + 4/60 + 3/60 = 12/60 = 1/5$. So $2 \times (A+B+C \text{ rate}) = 1/5 \rightarrow (A+B+C) \text{ rate} = 1/10 \rightarrow$ together they finish in 10 days.

Q: Trap question: If 9 spiders make 9 webs in 9 days, how many days does 1 spider take to make 1 web?

A: 9 days — NOT 1 day. Each spider makes 1 web in 9 days regardless of how many spiders work in parallel (they're not collaborating on the SAME web). This tests whether you recognize the rate-per-individual stays fixed, the classic trap being to assume the answer scales down to 1.

Q: Method — pipes filling and draining together: Two pipes fill a tank in 30 and 40 minutes separately; a third pipe drains it in 24 minutes. With all three open, how long to fill the tank?

A: Combined rate = $1/30 + 1/40 - 1/24$ (subtract the draining pipe's rate). LCM=120: $4/120 + 3/120 - 5/120 = 2/120 = 1/60$. So the tank fills in 60 minutes.

Q: Method — provisions lasting fewer days with more people (inverse proportion again): 1,000 soldiers have 70 days of provisions. If 400 more soldiers join, how long will provisions last?

A: People $\times$ Days = constant (total provisions). $1000 \times 70 = 1400 \times x \rightarrow x = 70000/1400 = 50$ days.

Q: Method — find the second worker's solo time from a combined time and the first worker's solo time: A and B together finish a job in 16 days; A alone takes 48 days. How long does B alone take?

A: B's rate = $(A+B)$'s rate - A's rate = $1/16 - 1/48 = 3/48 - 1/48 = 2/48 = 1/24$. So B alone takes 24 days.

Q: Trap question: If 5 persons complete 5 projects in 5 days, how long do 50 persons take to complete 50 projects?

A: Still 5 days — NOT 50 days. The ratio of people to projects stays the same (1:1), so the per-project time doesn't change; 50 people working on 50 projects is just 10 independent copies of the original 5-person/5-project scenario happening in parallel.

MOCK TEST (WITH ANSWERS)

1. 15 men complete a work in 24 days. In how many days will 20 men finish the same work?

- A. 16 days
- B. 18 days ✓**
- C. 20 days
- D. 30 days

Explanation: $15 \times 24 = 20 \times x \rightarrow x = 360/20 = 18$ days. [easy, authored]

2. A and B together can finish a job in 18 days; A alone takes 45 days. How long does B alone take?

- A. 24 days

- B. 27 days
- C. 30 days ✓**
- D. 36 days

Explanation: B's rate = $\frac{1}{18} - \frac{1}{45} = \frac{5}{90} - \frac{2}{90} = \frac{3}{90} = \frac{1}{30}$. B alone takes 30 days. [medium, authored]

3. If 7 workers build 7 walls in 7 days, how many days does 1 worker take to build 1 wall?

- A. 1 day
- B. 7 days ✓**
- C. 14 days
- D. 49 days

Explanation: 7 days — each worker's individual rate is unaffected by how many work in parallel on separate walls. [medium, authored]

4. Two pipes fill a tank in 12 and 15 minutes separately; a third pipe drains it in 20 minutes. With all three open, how long to fill the tank?

- A. 8 minutes
- B. 10 minutes ✓**
- C. 12 minutes
- D. 15 minutes

Explanation: Rate = $\frac{1}{12} + \frac{1}{15} - \frac{1}{20}$. LCM=60: $\frac{5}{60} + \frac{4}{60} - \frac{3}{60} = \frac{6}{60} = \frac{1}{10}$. Time = 10 minutes. [medium, authored]

5. 800 soldiers have provisions for 60 days. If 200 more soldiers join, how long will the provisions last?

- A. 40 days
- B. 45 days
- C. 48 days ✓**
- D. 50 days

Explanation: $800 \times 60 = 1000 \times x \rightarrow x = 48000 / 1000 = 48$ days. [easy, authored]

6. A, B, and C can do a job alone in 24, 30, and 40 days respectively. Working together, how long will they take?

- A. 8 days
- B. 10 days ✓**
- C. 12 days
- D. 15 days

Explanation: Combined rate = $\frac{1}{24} + \frac{1}{30} + \frac{1}{40}$. LCM=120: $\frac{5}{120} + \frac{4}{120} + \frac{3}{120} = \frac{12}{120} = \frac{1}{10}$. Together they take 10 days. [medium, authored]

7. If 8 people complete 8 tasks in 8 days, how long do 40 people take to complete 40 tasks?

- A. 8 days ✓**
- B. 20 days
- C. 40 days
- D. 320 days

Explanation: Still 8 days — the people-to-task ratio (1:1) is unchanged, so per-task time is unchanged. [medium, authored]

Part C – General English: Poem: "The Secret of the Machines" — Rudyard Kipling

READING MATERIAL

A poem narrated FROM THE PERSPECTIVE of machines themselves, describing how they're made, what they need to run, their capabilities, and their fundamental limitation: they're powerful but lifeless without humans. Real coaching-site evidence tests figure-of-speech identification heavily.

Q: Who wrote "The Secret of the Machines", and what is its central message?

A: Rudyard Kipling. Central message: machines are spoken of in first person ('we'), describing their manufacture and impressive capabilities (working 24 hours, doing tasks humans do), but the poem's real 'secret' is that machines are entirely dependent on humans to create and direct them — without people, they are worthless. The poem argues we should use machines to empower ourselves, not become dependent on them.

Q: How does the poem describe the manufacturing of machines?

A: Raw materials (iron-ore, minerals) are taken from ore-beds and mines, melted in furnaces, then hammered and shaped into various designs, cut and measured to the required size and weight.

Q: What do the machines say they need to operate, and what are their capabilities?

A: Just water, coal, and oil. They can work 24 hours a day, and can perform tasks that replace or assist human work (reading, writing, counting, pulling, hauling, pushing, lifting, driving).

Q: What is the central LIMITATION of machines, per the poem?

A: They cannot feel emotions like humans — they can only obey commands, and can be dangerous if handled improperly. Ultimately, however capable, they remain mere creations of the human brain.

Q: Identify the figure of speech: "We were taken from the ore-bed and the mine, we were melted in the furnace and the pit."

A: Anaphora — repetition of 'we were' at the start of successive clauses.

Q: Identify the figure of speech: "Some water, coal, and oil is all we ask!"

A: Assonance — repetition of vowel sounds within the line.

Q: Identify the figure of speech: "We will serve you four and twenty hours a day!"

A: Hyperbole — an exaggerated, emphatic claim about constant availability.

Q: What is the rhyme scheme of the poem's stanzas (e.g. ask/play/task/day)?

A: ABAB.

MOCK TEST (WITH ANSWERS)

1. "The Secret of the Machines" was written by:

- A. Walter de la Mare
- B. Rudyard Kipling ✓**
- C. Robert Frost
- D. John Masefield

Explanation: Rudyard Kipling. [easy, web-research]

2. According to the poem, what is the central LIMITATION of machines?

- A. They break down too often
- B. They cannot feel emotions and merely obey commands ✓**
- C. They are too expensive to build
- D. They require constant supervision by experts

Explanation: They cannot feel emotions, only obey commands. [medium, web-research]

3. "We were taken from the ore-bed and the mine, we were melted in the furnace and the pit" uses which figure of speech?

- A. Apostrophe
- B. Anaphora ✓**
- C. Personification
- D. Metaphor

Explanation: Anaphora — repeated 'we were' structure. [medium, web-research]

4. "We will serve you four and twenty hours a day!" is an example of:

- A. Alliteration
- B. Repetition
- C. Hyperbole ✓**
- D. Personification

Explanation: Hyperbole. [medium, web-research]

5. What do the machines, in the poem, say they need to operate?

- A. Electricity and steel only
- B. Water, coal, and oil ✓**
- C. Sunlight and wind
- D. Human supervision only

Explanation: Water, coal, and oil. [easy, web-research]

6. The rhyme scheme of the poem's stanzas (e.g. ask/play/task/day) is:

- A. AABB
- B. ABBA
- C. ABAB ✓**
- D. No fixed rhyme

Explanation: ABAB. [medium, web-research]